
Grounding Image Captions: Reducing Hallucination in Vision-Language Models

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1 Implementation Summary

The question of this project is whether vision-language models mention objects not present in the image, and whether we can reduce this through better caption selection. Our project combines two papers to address this problem.

1.1 BLIP

BLIP is a vision-language model that generates natural language captions by learning joint representations between visual features and textual tokens. It uses a CapFilt mechanism to clean noisy training data. We use the official pretrained weights `model_base_caption_capfilt_large.pth` released by Salesforce Research. BLIP was pretrained on datasets including COCO and related vision-language corpora, making COCO a training-aligned evaluation dataset. The BLIP paper recommends nucleus sampling for diverse caption generation. Sampling-based decoding is used to generate diverse candidate captions, which is consistent with prior work such as BLIP where sampling is used to produce varied captions in data generation pipelines.

1.2 CHAIR

CHAIR (Rohrbach et al. 2018) directly measures how often captions mention objects not present in the image. It defines two scores, CHAIRs which is the fraction of captions with at least one hallucinated object at the sentence level and CHAIRi which is the fraction of all object mentions that are hallucinated at the instance level. Lower is better for both. Together they give a complete picture of how widespread and how severe hallucination is.

An important design decision from the CHAIR paper is that ground truth must combine two annotation sources: instance segmentation labels and objects extracted from human reference captions. Using either source alone underestimates or overestimates hallucination. The paper also identifies language prior bias as the root cause of hallucination models add objects that commonly co-occur with objects actually present in the scene, even when those objects are not visible.

1.3 Our Approach

We use BLIP as the captioning model and CHAIR as the evaluation tool. Our contribution is a hybrid caption selection mechanism instead of taking a single greedy caption, we generate a pool of six captions per image and select the most visually grounded one using a hybrid scoring function that combines CLIP visual alignment object count penalty and length penalty.

2 Replication Results

2.1 Setup

We evaluate on 500 MS COCO val2017 images for Milestone 2, with the full 5000-image run reserved for Milestone 3. Ground truth is constructed in a CHAIR-style manner by combining segmentation annotations from `instances_val2017.json` and objects extracted from reference captions in `captions_val2017.json`, following the approach described in the original CHAIR paper. We use the official synonym list from the CHAIR paper authors repository (LisaAnne/Hallucination) which maps 403 natural language expressions to COCO category names such as *man* to *person*, *bike* to *bicycle*, and *fridge* to *refrigerator*. Words are singularized using the `inflect` library before matching.

The following output shows how segmentation and caption sources are combined for three random images, demonstrating how each source contributes objects the other missed.

```
... Images in combined GT: 5000
Image: 259690
  seg GT   : {'truck', 'baseball bat', 'chair', 'person', 'bench', 'bus', 'car'}
  caption GT: {'baseball bat', 'person', 'sports ball', 'chair'}
  combined  : {'truck', 'baseball bat', 'chair', 'person', 'bench', 'bus', 'sports ball', 'car'}

Image: 284279
  seg GT   : {'bird'}
  caption GT: {'bird'}
  combined  : {'bird'}

Image: 421060
  seg GT   : {'person', 'skis'}
  caption GT: {'person'}
  combined  : {'person', 'skis'}
```

Our implementation follows the CHAIR paper methodology but may not exactly replicate the original evaluation script behavior. We therefore refer to our metric as CHAIR-style evaluation throughout this report and plan to refine it closer to the original protocol in Milestone 3.

The BLIP paper uses beam search with beam size 3 for caption generation. We attempted this but encountered a tensor size mismatch error in the BLIP open-source codebase when using single-image batch size with beam search. We use greedy decoding as the baseline instead. Since both baseline and our proposed method use the same decoding strategy, all comparisons remain valid.

2.2 Baseline Caption Generation

We generate one greedy caption per image for all 500 selected images using `sample=False`, `num_beams=1`, `max_length=20`, `min_length=5`. The following output shows generated captions at every 50-image interval, confirming that BLIP produces coherent and varied captions across diverse scenes including people, animals, vehicles, and indoor settings.

```
... 50/500 -- a man sitting on the floor taking a picture of himself in a mirror
100/500 -- two men loading luggage into a car at an airport
150/500 -- a bus is parked on the side of the road
200/500 -- two brown bears walking on a dirt road
250/500 -- a cat is standing on a shelf in front of a television
300/500 -- a man and a woman holding wine glasses
350/500 -- a person in a red jacket skiing down a snowy hill
400/500 -- a young girl riding on the back of a horse
450/500 -- a bowl of oranges sitting on a table
500/500 -- a bird sitting on a branch with a tree in the background
Done: 500 captions generated.
```

2.3 Baseline CHAIR-style Results

Running CHAIR-style evaluation on all 500 baseline captions confirms that BLIP hallucinates objects in a measurable fraction of captions. The following output shows the evaluation result.

```
... Baseline CHAIR results:
CHAIRs: 5.0 %
CHAIRi: 3.29 %
hallucinated captions: 25 / 500
hallucinated objects: 27 / 821
```

These results confirm the core finding of Rohrbach et al. (2018): captioning models hallucinate objects and this cannot be detected by CIDEr or BLEU. BLIP at CHAIRs = 5.0% performs better than the older models in the original CHAIR paper which ranged from 7.4% to 17.7%, consistent with BLIP being a more recent and capable model.

Looking at the specific hallucinated captions reveals the language prior bias the CHAIR paper describes. BLIP added *couch* when it saw teddy bears, *microwave* when it saw a kitchen with a stove and refrigerator, and *cup* when it saw a sandwich. In each case the model applied a learned co-occurrence pattern from training data rather than strictly grounding its output in what is visible.

3 Findings and Challenges

3.1 Finding: Sampled Captions Vary in Hallucination Rate

A key observation that motivates our approach is that nucleus-sampled captions show meaningful variance in hallucination rate. We generated five nucleus-sampled captions per image (top_p=0.9, seed=42 for reproducibility) and ran CHAIR-style evaluation independently on each sample position across all 500 images. The following output shows example sampled captions, illustrating how the same image produces meaningfully different captions across samples with varying levels of specificity.

```
... 50/500
  a man that is sitting in front of a mirror
  there is a man that is taking a picture in the mirror
  a man sitting in front of a mirror
  the man is taking a picture of himself in the mirror
  a man sitting on the floor taking a photo
100/500
  a pair of men loading luggage from an open car door
  two men pushing a luggage cart at the airport
  people standing around luggage with cars parked in the background
  two men loading luggage into an suv at a baggage claim
  two people pushing luggage towards the trunk of an suv
```

The following output shows the hallucination rate per sample position. CHAIRs ranges from 3.6% to 6.6% with a variance of 1.0624.

```
... sample 1: CHAIRs=6.0% CHAIRi=4.31%
sample 2: CHAIRs=6.6% CHAIRi=4.65%
sample 3: CHAIRs=5.2% CHAIRi=3.61%
sample 4: CHAIRs=3.6% CHAIRi=2.51%
sample 5: CHAIRs=4.8% CHAIRi=3.62%

variance: 1.0624
min CHAIRs: 3.6 %
max CHAIRs: 6.6 %
baseline CHAIRs: 5.0 %
```

The best sample position outperforms the greedy baseline, proving that better captions exist in the pool. The worst position is substantially worse, showing we cannot simply pick any random sample. An intelligent selector is needed and this directly motivates our hybrid selection approach.

3.2 Challenges Encountered

Beam search error. The BLIP paper uses beam search with beam size 3. We encountered a `RuntimeError` due to tensor size mismatch in the BLIP codebase with single-image batch size. We use greedy decoding as the baseline. The comparison remains valid since both methods use identical decoding.

Synonym map loaded zero entries. Our initial implementation reported zero synonym entries loaded from the official CHAIR file. The root cause was a format mismatch. We expected lines of the form `category: synonym1, synonym2` with a colon separator, but the actual format has no colon and the first word on each line is the COCO category name. Fixing the parser loaded 403 correct entries. Without this fix, CHAIR-style scores were systematically underestimated because common words like *woman*, *bike*, and *fridge* were not being matched to their COCO categories.

Top-K changed to nucleus sampling. Our Milestone 1 plan proposed Top-K beam search for generating multiple candidate captions. Since beam search caused the error above, and the BLIP paper itself recommends nucleus sampling with `top_p = 0.9` for diverse caption generation, we adopted nucleus sampling. This is an informed revision that is better supported by both the codebase and the original paper.

4 Value Additions

4.1 Hybrid Caption Selection

We compare three caption selection strategies, all applied to the same BLIP model. Baseline uses greedy decoding with no selection. CLIP-only selection generates a pool of six captions per image one baseline greedy caption plus five nucleus-sampled captions and picks the one with the highest CLIP cosine similarity score. Our hybrid method extends CLIP-only by adding two penalty terms:

$$score = \alpha \cdot CLIP_score - \beta \cdot object_penalty - \gamma \cdot length_penalty \quad (1)$$

where $\alpha=1.0$, $\beta=0.3$, $\gamma=0.2$. The CLIP score measures visual alignment between image and caption as the normalized cosine similarity between CLIP image and text embeddings. The object penalty discourages captions that mention many objects since more objects means more hallucination risk. The length penalty discourages overly specific captions that make more claims than necessary. The baseline greedy caption is always included in the pool so that selection can never perform worse than the unmodified BLIP output. All three methods are evaluated using the same CHAIR-style metric on the same 500 COCO images, making the comparison completely fair.

The following output shows the scoring breakdown for one example image, illustrating how each component contributes to the final selection decision.

```

... Image: 00000000139.jpg
GT objects: {'dining table', 'book', 'chair', 'clock', 'microwave', 'person', 'potted plant', 'vase', 'refrigerator', 'tv'}

baseline:
caption : a woman standing in a living room with a television
CLIP    : 0.6409
obj pen : 0.2
len pen : 0.0
hybrid  : 0.5809

sample 1:
caption : a living room with hardwood floors and an orange wall
CLIP    : 0.6459
obj pen : 0.1
len pen : 0.0
hybrid  : 0.6159

sample 2:
caption : a living room with a fire place and a woman in the window
CLIP    : 0.6384
obj pen : 0.1
len pen : 0.15
hybrid  : 0.5784

sample 3:
caption : a lady in an apartment watching a television set
CLIP    : 0.6374
obj pen : 0.2
len pen : 0.0
hybrid  : 0.5774

```

The following output shows the selected captions from each method across 500 images, illustrating how hybrid consistently selects shorter and more focused descriptions compared to CLIP-only.

```

... 50/500
baseline : a man sitting on the floor taking a picture of himself in a mirror
CLIP     : a man sitting on the floor taking a picture of himself in a mirror
hybrid   : a man sitting in front of a mirror

100/500
baseline : two men loading luggage into a car at an airport
CLIP     : two men loading luggage into an suv at a baggage claim
hybrid   : two men pushing a luggage cart at the airport

150/500
baseline : a bus is parked on the side of the road
CLIP     : the bus is parked outside of the big building
hybrid   : the bus is parked outside of the big building

200/500
baseline : two brown bears walking on a dirt road
CLIP     : two brown bears walking on a dirt road
hybrid   : two brown bears walking on a dirt road

```

4.2 Results

The following outputs show the CHAIR-style evaluation results for all three methods and a bar chart comparison.

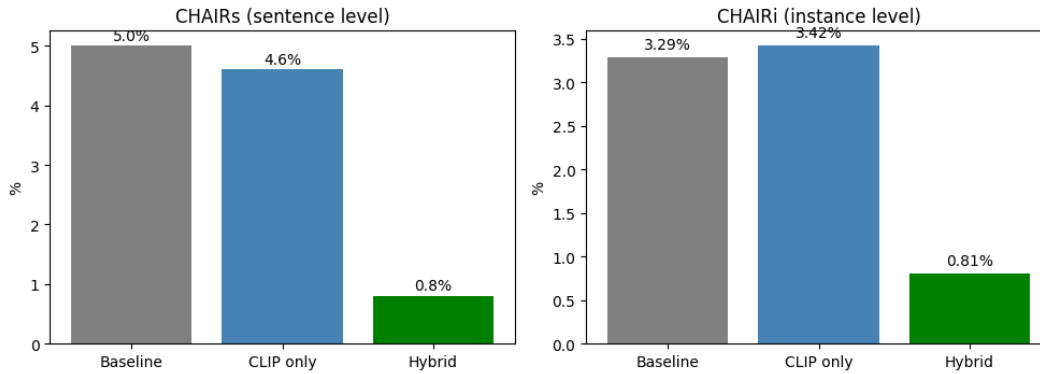
```

... Results on 500 COCO images:

Baseline  CHAIRs=5.0%  CHAIRi=3.29%
CLIP only CHAIRs=4.6%  CHAIRi=3.42%
Hybrid    CHAIRs=0.8%  CHAIRi=0.81%

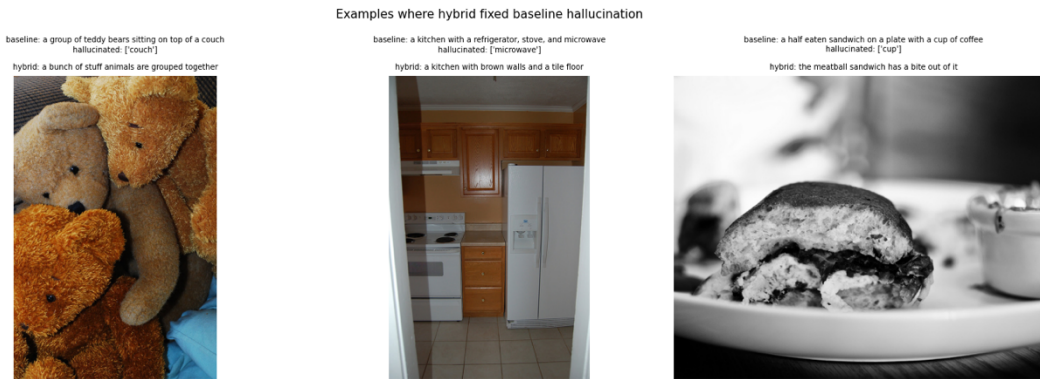
```

Baseline vs CLIP-only vs Hybrid - COCO val2017 500 images



Hybrid selection reduces hallucinated captions from 25 to 4 on the 500-image subset, corresponding to an 84% reduction in CHAIRs from 5.0% to 0.8%. CLIP-only selection reduces CHAIRs by only 0.4%, showing that the object count and length penalty terms contribute meaningfully beyond visual alignment alone.

The following shows three qualitative examples where baseline hallucinated but hybrid did not.



4.3 Diagnostic Analysis

The following output shows the diagnostic comparison including CHAIRs, CHAIRi, average caption length, and average object mentions per method.

Method	CHAIRs	CHAIRi	AvgLen	AvgObj
Baseline	5.0%	3.29%	9.61	1.64
CLIP only	4.6%	3.42%	10.28	1.52
Hybrid	0.8%	0.81%	9.24	0.98

Hybrid reduces average object mentions by 40% while CHAIRs reduces by 84%. This raises an important question, is hybrid genuinely selecting more visually grounded captions, or is it simply choosing shorter and simpler captions with fewer opportunities to hallucinate? The fact that the CHAIRs reduction is proportionally much larger than the object mention reduction is encouraging and suggests hybrid may be doing more than just being conservative. However we cannot confirm this without further analysis and we treat it as an open question for Milestone 3, where we plan an ablation study on the penalty weights.

4.4 Flickr30k Evaluation Plan

In Milestone 3 we plan to evaluate BLIP caption performance on Flickr30k as an out-of-distribution dataset. The purpose of this evaluation is simply to observe how the model behaves on images it has never seen during training. Since BLIP was pretrained on COCO-aligned data, Flickr30k represents a different visual distribution and gives us a sense of whether caption quality changes on unseen data. This is purely an observational exercise and not a validation of our hybrid method. The effectiveness of our hybrid selection approach is already demonstrated on COCO using the CHAIR-style metric, which is the proper evaluation for our contribution. For Flickr30k we will report baseline CLIP score only as a supporting observation, since CHAIR evaluation is not applicable without instance-level object annotations. We acknowledge that comparing hybrid versus baseline on Flickr using CLIP score would be biased since CLIP is part of our hybrid scoring function, and we will not make that comparison.

5 Project Timeline

Completed in Milestone 2: BLIP pipeline setup, CHAIR-style evaluation implementation, combined ground truth construction, synonym loading bug fix, baseline evaluation on 500 COCO images, nucleus sampling with 5 captions per image, variation analysis, CLIP model integration, hybrid selection implementation, evaluation of all three methods, diagnostic analysis.

Remaining for Milestone 3: Scale evaluation to full 5000 COCO val2017 images. Refine CHAIR implementation closer to the original paper protocol. Run ablation study on hybrid penalty weights to understand whether improvement comes from visual grounding or conservative selection. Evaluate on Flickr30k and report baseline CLIP score as out-of-distribution performance observation. Prepare final NeurIPS-style report, organize GitHub repository and deliver final presentation.